Hoffman2 Basics

Outline

1. What is a cluster?
2. Tour of directories + basic unix commands
3. How to edit files in hoffman?
4. How to move files into/out of hoffman?
5. Submit a toy job to the scheduler

What is a compute cluster?

Your Computer

1 node
8 CPUs
8 GB memory

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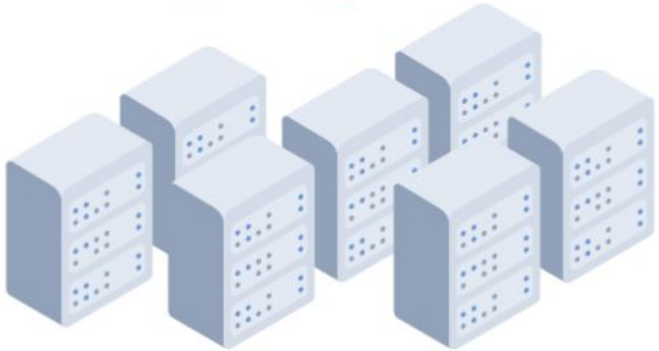
Hoffman 2

Computing Cluster



800 nodes
26k CPUs
179 TB

CC-by hutchdatascience.org

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An individual computer in the cluster

Your Computer

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8 CPUs
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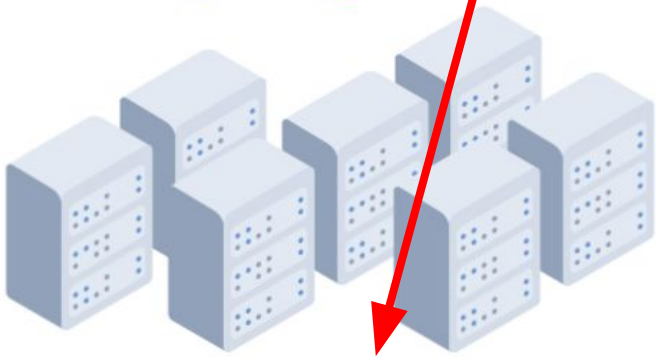
Hoffman 2

Computing Cluster



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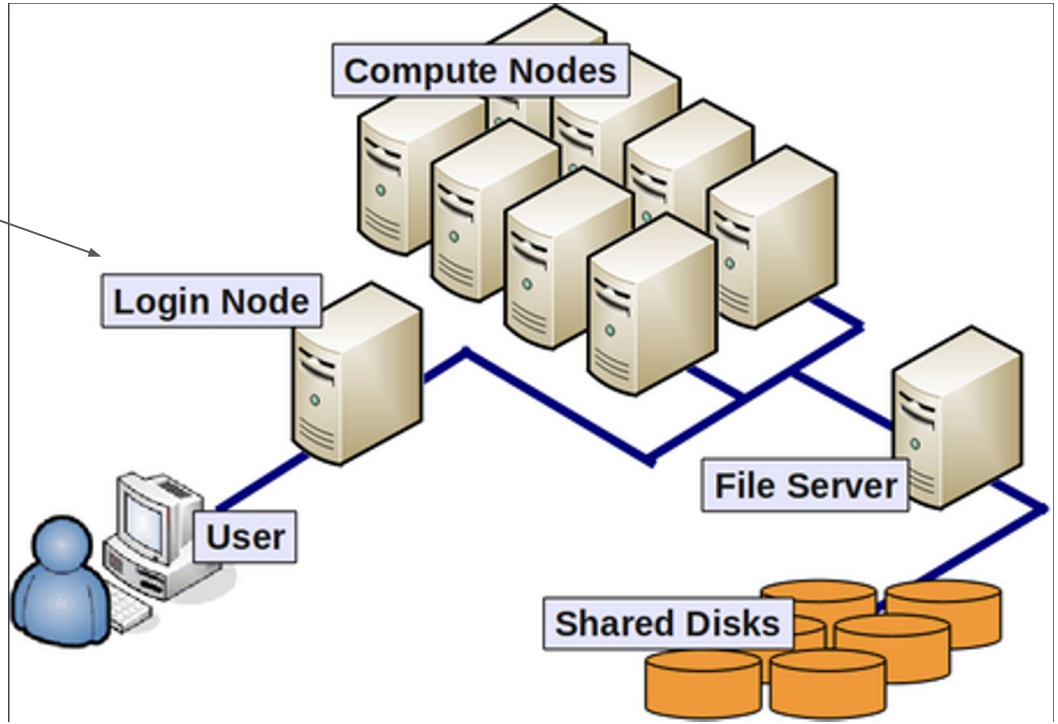
Where we enter ***we move away to do things***

We have memory allocated for us in the home directory on the login node but it's ***very*** limited



Where we do things or schedule them to be done

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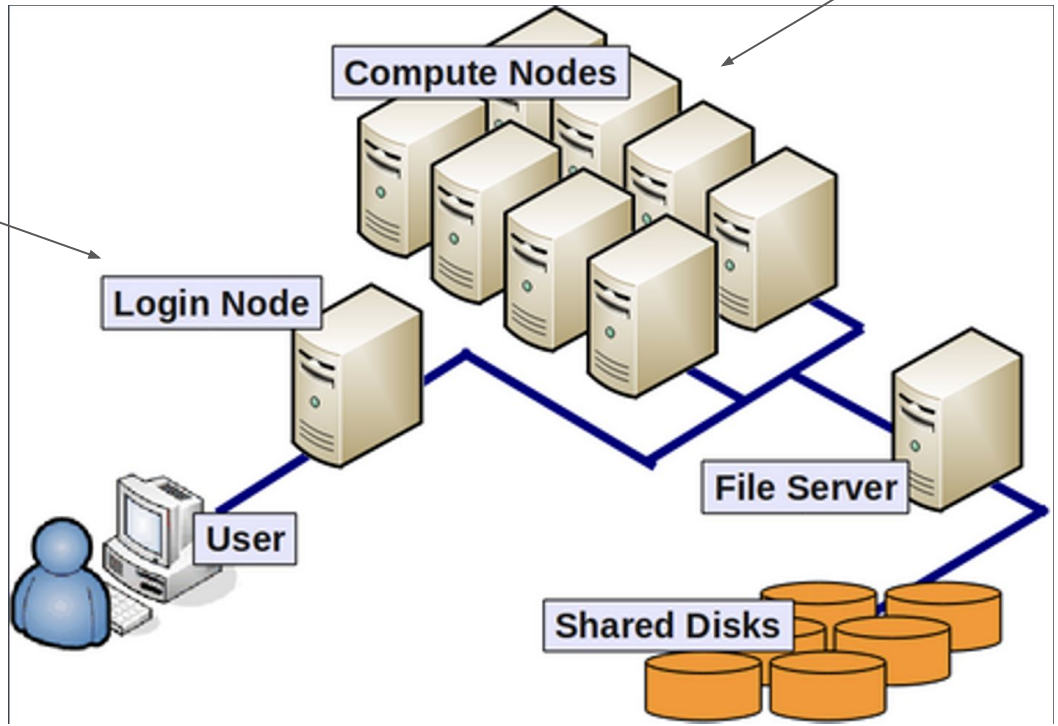

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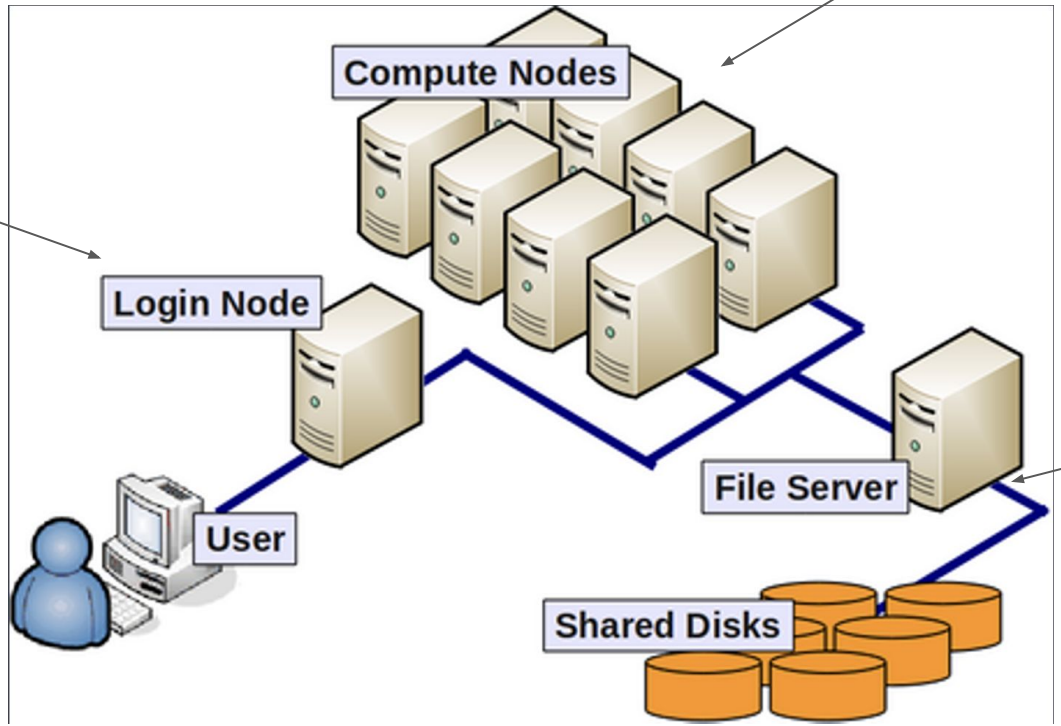

Computer with the primary purpose of providing a location to store data.

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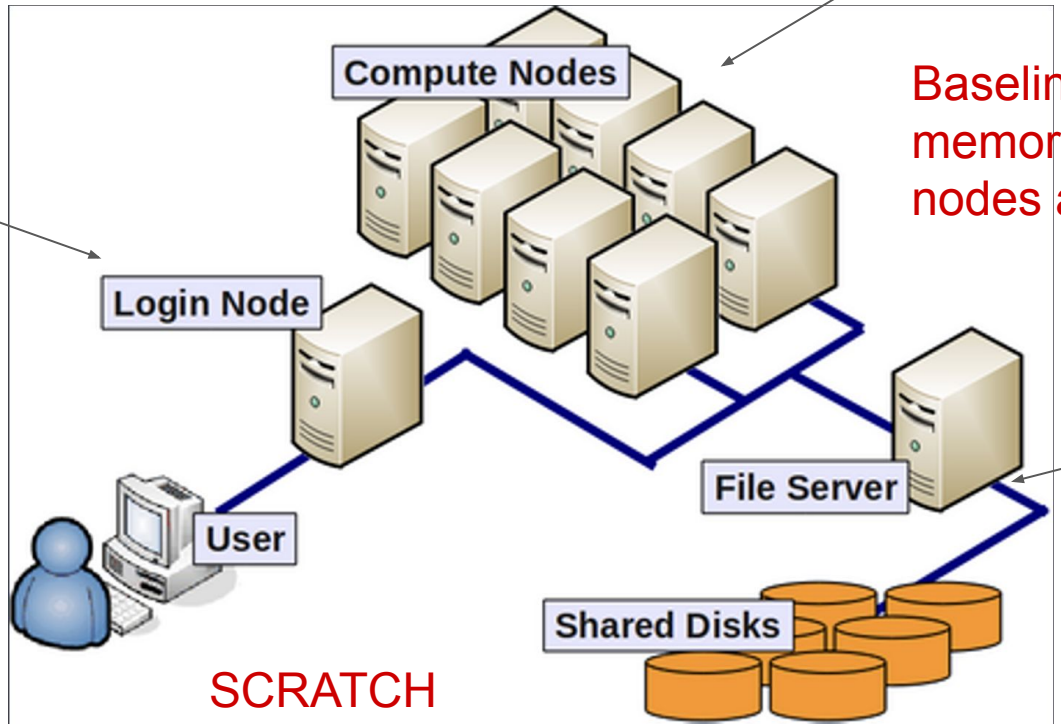


Baseline nodes and memory + Bigham Lab nodes and memory

Computer with the primary purpose of providing a location to store data.

SCRATCH
TEMPORARY 1TB

Log in

```
Open the terminal
```

```
ssh login_id@login1.hoffman2.idre.ucla.edu
```

Add this

Move out of login node into compute node

```
qrsh -l h_data=10G,h_rt=6:00:00
```

Basic Unix command architecture : [command] [where]

Example to **lis**t files in directory:

```
ls /u/home/e/eewade
```

Example **c**hange **d**irectory:

```
cd project-awbigham
```

Make file

```
touch  project-awbigham/testfile
```

Remove file

```
rm project-awbigham/testfile
```

Special shortcuts: `..` = directory behind current `.` = current directory `~` = home directory

Quick tour

```
[[eewade@login1 ~]$ ls
~                      core.12759         index.html    perl5                 project-klohmuel  start_rstudio_apptainer
annotate_windows.Rout  hmm_decoded.csv    index.html.1  project-awbigham      __pycache__       wqe.tar.gz
bigsummer.tar.gz       hmm_emissions.png  index.html.2  project-kirk          R
bin                    hmm_results.png    packages.rds  project-kirk-bigdata  rotations
[[eewade@login1 ~]$ ls $SCRATCH
gds  logs  processing  rstudiotmp
[[eewade@login1 ~]$ ls project-awbigham/
daily  data  rotation_daily.tar.gz  Software  submissionscript.sh
[eewade@login1 ~]$
```

Using pre-installed software

`module avail` — look at what’s available

`module load python` — load something

Viewing and editing files

Built in text editors:

nano , vim, emacs, etc.

RStudio Server IDE - https://github.com/Bigham-AnthroGenetics/Bigham-Lab-Tips-and-Tricks/blob/main/AccessingRStudioServer.md

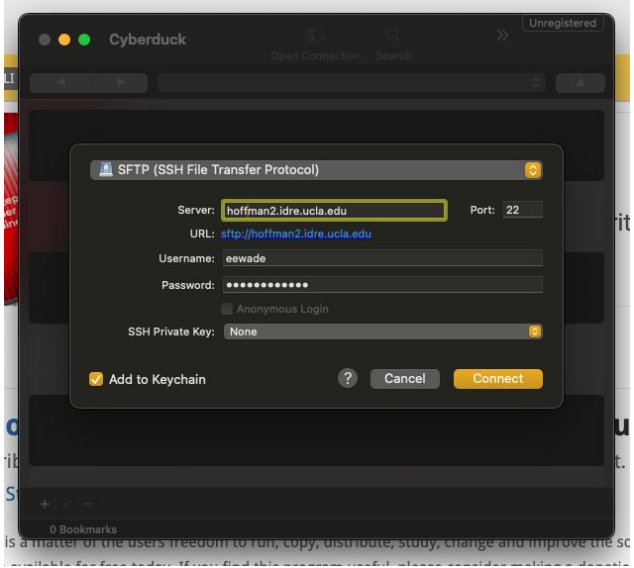
Moving files

From local to hoffman:

scp location/on/local eewade@hoffman2.idre.ucla.edu:/location/to/moveto/hoffman

Cyberduck

Globus



Connecting globus

Globus Connect Software (Local Desktop)

If you want to transfer files to or from your local desktop machine, you need to download and install the Globus Connect software (one-time). You will need to do this step on each of your desktop machines whose files you want to transfer using Globus.

1. Point your browser at http://www.globus.org and click Globus Connect. You will see a popup window in a web page. If you don't see the popup window, click the Get Globus Connect link on that page.
2. In Step One, click the button corresponding to your local platform (Mac OS, Linux or Windows) to download to your local desktop machine.
3. In Step Two, enter an Endpoint Name to identify your local machine in the Endpoint Name field, (you can ignore the Description field), and click Generate Setup Key.
4. Copy the setup key to some file and save it, since you will not see it again.
5. Install the downloaded Globus Connect software on your local desktop.
6. Run Globus Connect Installation (Windows: globus_connect_install.exe) for the initial setup, and paste in the setup key when prompted.
7. You should now be able to transfer files by following the directions below.

Transfer - Local to Cluster

How to transfer files between the Hoffman2 Cluster and your local computer

1. If you want to transfer files to or from your local desktop machine, start Globus Connect (Windows: gc.exe, Mac: Applications/Globus) on your local machine. A small status window will appear. When the connection has been made, the dot at the left of the connection will turn green. (MAKE SURE THE KEY IS ADDED - Instructions Above.
2. Point your browser at http://www.globus.org, click Sign In, and click Transfer Files. A web page with two Endpoint fields will display.
3. In one Endpoint field, pull down the expand menu and select a site. If you are running Globus Connect, the first name in the list is your local desktop machine. When you select your local desktop machine, a list of your home directory files and directories will be displayed.
4. In the other Endpoint field, either pull down the expand menu and select hoffman2#ucla, or type in hoffman2#ucla.
5. When you select hoffman2#ucla a popup window will ask you for your myproxy server username and passphrase. Enter your Hoffman2 username and its password. Leave the Server DN field as it is. The default Lifetime value is 12 hours. If you are transferring a large amount of data, you may need to increase the Lifetime value. Click the Authenticate button.
6. Your Hoffman2 Cluster home directory and its contents will display.
7. To transfer files between endpoints, select a file or directory from each list, then click one of the large arrow buttons to tell Globus the desired direction of the transfer.

You will receive an automatic email from Globus Notification (notify@globus.org) when the file transfer has completed. To have Globus show you the status and history of your file transfers, from its Go To pull-down menu, select View Transfers.

https://www.ccn.ucla.edu/wiki/index.php/Hoffman2:Using_Globus

Sending a job to the scheduler

```
#### submit_job.sh START ####
#!/bin/bash
#$ -cwd
# error = Merged with joblog
#$ -o joblog.$JOB_ID
#$ -j y
## Edit the line below as needed:
#$ -l h_rt=1:00:00,h_data=1G
## Modify the parallel environment
## and the number of cores as needed:
#$ -pe shared 1
# Email address to notify
#$ -M $USER@mail
# Notify when
#$ -m bea

# echo job info on joblog:
echo "Job $JOB_ID started on:   " `hostname -s`
echo "Job $JOB_ID started on:   " `date `
echo " "

# load the job environment:
. /u/local/Modules/default/init/modules.sh
## Edit the line below as needed:
module load gcc/4.9.3

## substitute the command to run your code
## in the two lines below:
echo '/usr/bin/time -v hostname'
/usr/bin/time -v hostname

# echo job info on joblog:
echo "Job $JOB_ID ended on:   " `hostname -s`
echo "Job $JOB_ID ended on:   " `date `
echo " "
#### submit_job.sh STOP ####
```

```
qsub submit_job.sh
```

https://www.hoffman2.idre.ucla.edu/Using-H2/Computing/Computing.html#submitting-batch-jobs

Sending a job to the scheduler

Array Job submission script

```
#### submit_arrayjob.sh START ####
#!/bin/bash
#$ -cwd
# error = Merged with joblog
#$ -o joblog.$JOB_ID.$TASK_ID
#$ -j y
## Edit the line below as needed:
#$ -l h_rt=200,h_data=50M
## Modify the parallel environment
## and the number of cores as needed:
#$ -pe shared 1
# Email address to notify
#$ -M $USER@mail
# Notify when
#$ -m bea
#$ -t 1-49:1

# echo job info on joblog:
echo "Job $JOB_ID.$SGE_TASK_ID started on:   " `hostname -s`
echo "Job $JOB_ID.$SGE_TASK_ID started on:   " `date `
echo " "

# load the job environment:
. /u/local/Modules/default/init/modules.sh
## Edit the line below as needed:
#module load gcc/4.9.3

## substitute the command to run your code
## in the two lines below:
echo '/usr/bin/time -v ./add_by_component.sh'
/usr/bin/time -v ./add_by_component.sh

# echo job info on joblog:
echo "Job $JOB_ID.$SGE_TASK_ID ended on:   " `hostname -s`
echo "Job $JOB_ID.$SGE_TASK_ID ended on:   " `date `
echo " "
#### submit_arrayjob.sh STOP ####
```

Useful for chromosome split

```
qsub submit_arrayjob.sh
```

https://www.hoffman2.idre.ucla.edu/Using-H2/Computing/Computing.html#submitting-batch-jobs

Let’s do a task

1) Get phased vcf from 1KG
2) Make plink files for chromosome 22 in a job submitted to the cluster
    a) Will grab script from github

Walk through is here:
https://github.com/Bigham-AnthroGenetics/Bigham-Lab-Tips-and-Tricks/blob/main/labmeeting/examplewalkthrough.md

Adapted from Kelsey Workshop - thank you Emma Y